

ActInf Textbook Group ~ Cohort 5 ~ Session 5 (Chapter 2, Part 2) ~ 10/17/2

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 5 (Chapter 2, Part 2) ~ 10/17/2023 | 1:01:35

all right welcome back everyone it is
October
17th and we're in our second discussion
on chapter
two so
let's head over there and
um first does
anybody interesting little colors were
added does anyone want
to give
any opening thoughts or
Reflections on chapter two or just bring
up some
question so I could
share
briefly comparison to
the previous
sections uh it took me longer for
sure and I had to do more uh search for
more reference material because I think
it's the first section that's
really trying to introduce the the
mathematics um of the
framework
and uh the the first Parts where we're
discussing uh very free energy which is
only taking into account information
about the past and the present
to um to compute uh free
energy
um this was relatively
straightforward um I think the section
on expected free energy is a more
significant
leap and I think I need to spend more

time I'm

um uh analyzing that and and really

making sense

of of that part of of the

chapter awesome thank

you anyone else want again just any part

of two that stuck with him or a question

they have or a a thought on the role of

the low rad

um I will just uh Echo the sentiment

that um I I think I I finally you know

um got the handle on free energy as a

concept but the expected free energy is

still completely impenetrable making

that that that um uh cognitive leap is

is is is still a little bit beyond

beyond my

Reach cool we can definitely look over

it okay at the at the uh at the very

small font

overview we have perception as inference

so it's still

passive

is even though we can discuss there are

some ways of action is related to even

what might be seen as passive inference

background on basian statistics

example on the background on basian

statistics which we can look at again

but there's an object in a person's hand

and they're updating their beliefs about

what they think that object is based

upon what they observe it might be an

apple it might be a frog and then they

see it jump and they update their belief

from thinking that it's probably an

apple to probably a frog

um that case is carried through in the

exact Bayesian setting and for smaller
State spaces and smaller scale
computational problems you can utilize
exact

base however for larger State spaces you
can't always do exact Bay and so that
requires an approximation method and
variational free energy or the evidence
lower bound is a common approximation
method for doing variational Bayesian
inference

um doing approximate Bayesian inference is
then mobilized in this particular
partition cybernetic

setting which leads to a discussion of
action still we're in

the um cybernetic setting continuing to
discuss like perception and action as
part of this unified approximation
challenge

then we get to the real

equations equation 2.5 and 2.6 are like
the two key equations in this chapter

2.5 is the variational free energy so
here is where we have um the ability to
build on people's work previously and
also make massive massive um
contributions with our studying here in

2.

five previous cohorts have used the
ontology to provide natural language
descriptions of the

equation so if this just looks like a
lot of

symbolology then it may be helpful to
read the ontology description of

it 2.5 variational free energy is about
beliefs and data so variational free

energy is like the real
time um sense making Vibe check it's
very closely related to how surprising
incoming observations are given
beliefs some more discussion on how the
kale diversions comes into
play and then it goes from consideration
of the real-time variational free energy
into the future oriented expected free
energy and planning as inference with
equation

2.6 but that's not something that all
cognitive systems necessarily do not all
systems engage in explicit planning and
then there's some discussions about
special cases of expected free energy
and that's the end of the low road so
chapter 2 pretty much starts with a
simple exact Baye

setting slides over to the approximate
variational case applies it to
cybernetics and then introduces the two
key equations in 2.5 and

2.6 variational and expected free
energy just to kind of give an overview
but but like so as as you have
highlighted like those are the exact
things to pick up in the low
road

so anyone else want to just bring up any
thought or question or what what area
can we go
to yeah I think maybe before we move on
to the more
technical uh

details I think it's important
to to highlight the significance of this
jump like it's a a conceptual

leap that was very striking to me and I think after you spent some time um studying this kind of area it's it becomes kind of intuition but but this idea of perception as inference um in contrast to what at least throughout my life the way that I've thought about seeing things has been this the straight transction from from objects in the world to uh just lights on my retina and then some processing processing through my brain and then just kind of this image being projected somewhere um but the the predictive aspect the the inferential top down component of it is is I think it's a remarkable idea and it's interesting that um this isn't it's not really something that you can tell at least I I have couldn't couldn't tell just by observing how I look at the world at least not not earlier in my life so so I'm I just wanted to highlight that as a as an interesting uh but significant like a significant leap I think in in thinking about perception yes Sarah and then anyone else hello I have a disclosure a comment and a question so my disclosure is I read the wrong chapter I missed two weeks so I thought I'd missed two chapters but obviously I didn't uh I did read this chapter but I read it like six months ago so I'm just

flicking through very fast now so that
my disclosure my comment is yeah I
totally get the idea it's such a it's
such like a a flip to go from this idea
of light coming in your eyes I remember
in primary school being told off for
doing it the wrong way feeling like
light goes out and she's like no it goes
in and like we we kind of grow up with
this idea like we're sensory machines
but then I feel like once you make the
flip it's all predictive you it's really
actually hard to go back and like not
see it anymore and then my question was
so is I guess it does relate to this
chapter which is lucky because you know
that's what we're actually on so like
figure 2.5 right so You' got this idea
of the perception and the action and one
thing I've been struggling with is the
how they're different right because
perception changing your perception is
actually an internal action so it's
you're updating a model well what's that
that's changing a synaptic way that's an
action but it's just being performed
internally so one thing I've didn't
realize I was struggling with this till
today is uh yeah how why is perception
separate to action and I know like the
idea is it's a unified theory of
perception action but why do we still
have those and how are they different am
I missing something so thank
you
yes
yeah okay H how how are perception and
action um similar and different and

related okay and

Kristoff um so I just want to say kind of couple things I like this idea of this what s just said about light light going out rather than and reminded me of a quote from uh kard that life has to be lived forwards but uh can only be understood backwards and there's there's something about the way we explain the world

um you know to ourselves or other is that has this kind of like inversion quality to it and and and um I kind of I can't put my finger on it precisely yet but I I see there's a kind of some kind of universal phenomenon that's that's lurking there but um about this leap from you know uh perception to to inference I remember one of the uh earlier papers by Carl Frist he mentioned something really fascinating and he mentions the hubel and wisel um uh experiments with um cat retina I don't know if you um if all of you are familiar with that but they basically were Shining Light um uh onto Cuts uh retina an aized cat and measuring response from some um um some neuron there for you know measuring electric response and they found these recept these ideas receptive fields that responded to various kinds of orientations and and and so on and the interesting phenomenon that they couldn't explain was end stopping so when you have a receptive field you know like it will it will respond to the specific neuron will respond to some

specific orientation and specific length
of the um of the rectangle that we're
projecting but when the rectangle gets
wide enough it stops
and the question was why and uh
Neuroscience couldn't quite explain this
but uh uh sort of you know predictive um
the sort of predictive framework um uh I
remember Carl mentioned that in one of
his papers it starts to suddenly makes
sense because kind of in nature you
don't really see short lines so when
you're projecting a when when you're
looking at the Horizon it's always a
very very long line but if it becomes
shorter then it's you know it's
something that um does not occur in
nature and therefore for the neuron
fires because it actually is a
prediction error so it's kind of
fascinating that uh um this this
inversion of this idea suddenly suddenly
made a phenomenon um you know uh
explainable which I thought was really
really
interesting yes
so first this uh like coming in and out
of the eye
um it is really a different view on
perception when as as um was brought up
like whether we think about well there's
this unrolling generative model that's
getting light touches from the sensory
information getting fine-tuned that's
the sort of like inside out view of
experience as opposed to a sensory
processing I mean sensory processing
even as a as a framework makes it sound

like the information is out there and we're going to like process the wheat separate the wheat from the chaff and then we're going to end up with like just the real you know we're going to have the pixels on the camera and then we're going to process the pixels and figure out whether it's a cat or a dog or um even further like we're going to process the pixels and then we're going to figure out what to do um as opposed to we're going to have this unfolding model about how things are and their likelihoods and and what to do and then we're going to get get just light touches from sensory data and um I guess the trip is that um if we believe that sensory processing is outside in the world will confirm our beliefs I think there's a few truth drops just in the visual experience that help us understand the Inside Out perception so the first is the blind spot which we don't attend to it's filled in perfectly even though there's a large um patch on the retina that doesn't get that much visual input second one is that in the periphery there are uh fewer um there's lower resolution vision and there's no color vision however in healthy Vision we have color and equal resolution and there's experiments that show like actually well it is lower resolution in the periphery but it's not attended to as such and then we just look over and then the third aspect is the Cade so multiple times per second our eyes are

moving and with active inference we can understand okay it's like we're when the Gaze is fixed our Precision is high and we're getting we're paying more attention to visual input and then right before we're ready to execute a cascade of course this is subconscious Precision is alleviated the eye moves and then it's like the the foot goes back on the gaps so the fact that the blind spot the periphery and the saccades are not attended to is is very um tangible that just our visual experience is what we generate it to be with of course information coming from what's external but it's not being processed merely from what's external okay so let's go to s's question here um how are action and perception similar different or related what's anything that that anyone wants to add on this or there's a few pieces to bring in the part about the planning um you know is that in between or is that part of action got great question well so um uh Kristoff brought up the Hubble and whizzle experiments with the cat retina and the selectivities um in different brain regions and that actually connects to one of the most cited uh Neuroscience papers certainly within the ACT INF space which is r and

Ballard

99 which provided some of the early strong evidence for predictive coding and so like well so in that retina experiment they had classical patterns like classical cell types oh this one responds to vertical lines this one responds to diagonal lines this one's horizontal or so on however there were these receptive

Fields like patterns of activation of visual neurons that were extra classical like they were not accounted for within a classical receptive field model and this

paper with empirical evidence and and mathematical

modeling established that predictive coding which is to say like expectations about visual input were able to account for some of these extra classical phenomena

um

Andrew oh yeah I just wanted to say just very generally on the the action perception relationship um active inference seems to take the stance that like we are actively inferring information around us and so rather than being like passive observers who are just sitting there we we engage in actions in order to um kind of impact our perception such as like you pick something up and look at it like you can look at it from different angles all of those are like actions that you're taking um that's that's my general sense of the relationship between the two as

opposed to some classical view that we're just you know sitting there we're taking in whatever the supposed reality is around us we're not actually doing anything we're just observers um that's it I don't know if that tracks yeah thank you um Jesse then

Manuel yeah I think I mean isn't the the two options either you change your mind or you change the world to minimize the free energy

so um that's kind of how I see it anything that you do that act that changes the World materially falls in the category of action where um changing your mind is perception I guess what I'm struggling with is that your mind is in the world right so you say you change your mind but that's the same thing as changing the world like I'm still struggling with the the difference because changing your mind it's just okay even even if we reduce it to it's changing firing rates it's changing patterns in in neurons it's not some esoteric mind concept it's still something physical we're changing there's a action there to change the the synchrony between different brain regions or whatever it is that causes that sort of perceptual switch but it's not like an esoteric concept it's a it's still something physical that's changing is what I'm struggling with and but I I completely get what you're saying like I do agree with that but then when you the smaller and smaller you get it's like what is changing your mind it's just a

changing weights between neurons it's
changing synchrony it's changing
something physical unless unless you
know maybe this will be the answer and
we'll find something metaphysical
finally in the in the brain
somehow awesome menw well and
then yeah just to give my two cents on
this point I think is really interesting
what s is saying but you know in the
formalism uh where the free energy
principle comes from my understanding is
you know you have to define a a
dynamical system which has many
variables and for this thing to apply
there has to be a concept of a mark of
blanket which will come later in the
book but it's like you have a separation
of variables that do not have direct
contact with other variables within the
system only through through this this
layer of
interaction and perception and action
are different there in the
directionality of the influences so
perception only influences the internal
state of the subsystem you care about uh
but doesn't do the opposite and or or
the sorry the the internal system
doesn't influence the perception States
and the opposite with action and when
you say yeah like the brain also is
taking an action in in a physical world
and that's true you know Freestone seems
to mention pretty often how these Mark
of blankets I think of them as you know
the mataska the Russian doll many layers
of these uh Mark of blankets on top of

each other so a cell it's doing its own
uh active inference but it's so it's had
its own Mark of blanket maybe defined by
its own cellular membrane and whatnot
but it's part of a circuit and that
circuit is also has now a more
functional Mark of blanket and it and
and they take uh take actions towards
other systems and and and you know many
many scales of this kind of thing but
the I think the more precise difference
is this of of what variables are
changing in your system versus what
variables you change of what is not the
system all defined by what the mark of
blanket is always in the in the sense
that there has to be something that is
that you care about and some and the
rest that you don't care about directly
yes very interesting and important point
which is a a simple or narrow technical
difference between action and perception
is for a given system of interest that
we're just going to focus on the inbound
dependencies are perceptual the outbound
dependencies are action and then with a
different
blanket or a different system of
Interest a given causal influence might
be under understood as well you know it
was an action for this for the speaker
but then it was a perception for The
Listener so then by taking uh uh by by
swapping a
perspective um we might give a different
AC INF onology tag to the same causal
Edge on a base
graph um raw and then

Andrew yeah I was thinking a bit about this uh after I got to figure 2.2 in the textbook um because they're the figures they're illustrating this um uh this interface or this this influence that uh an agent has uh on the World to uh generate some kind of uh observations why here and in my mind the difference between perception is in action is at least based on the formalism that we're given here is in the access that we have to the hidden States so out in the world there's there's some some process some generative process as they refer to it and and there's no way to observe though that uh the hidden states of that generative process and the hidden states are the causes of whatever uh observations you have so for example if you see an apple there's something out there in the world that is generating Apple like observations on your retina and then uh inside of your mind you have some also you have a generative model of what that uh process is in out in the world but you actually have some um maybe some influence some more direct influence over that uh over the configuration of the generative model and uh so even if you take some actions internally to update that model it's uh it's different than than like uh some proxy action that you take to change your observations out in the world um that's as far as the

distinction has have been clear enough
to me but I think this idea of a mark
off blanket which will probably Explore
More
um on in the high road to active
inference uh I think that's also a deep
an important idea
for establishing what it means to be uh
a system what it means to be a thing
that that is separate in some uh
significant way in some statistical
way perhaps from from the external
world thank you great Point even if we
chose to take a kind of active modeling
approach to internal action like through
attention or whether even conceptualized
of learning as action like just the
actual parameter updating of learning
not just like moving around in the world
only through the agent's boundary would
the object be intervened upon in the
world it'd be like you know if people
are watching a movie and they're they
might be doing intentional modulation or
something they're not intervening with
what's on the movie
screen versus an outgoing dependency
which is to say real over to
action then um does modify how hidden
States changed their
time

Andrew sorry um yeah know I think
everyone just before me largely Su up
what I wanted to say I guess I'll just
add um another way I kind of think about
it um is like the inclusion of the the
idea of preferences and the generative
model carries preferences um and in

differentiating the um the system of
Interest like the model versus the
process whose true States we don't have
direct access to and so on um like you
know for me a human is a generative
model like I have references or
homeostatic set points or however you
want to view those so I want to keep my
Hunger low I would like to get some good
sleep I would like to have a well
regulated body temperature and um while
I would say yes my body my mind and so
on are in fact part and parcel with the
world I'm in the world I don't think
that the world around me has the same
preferences I do I don't think it's
attempting to uh regulate my body and uh
maintain my home ostatic set points I
have to engage in in action as well as
like attempting to get rather tractable
accurate perceptions of the World Around
Me In order to act and and uh realize my
preferences that are you know all of
these
homeostatic um set points um active
inference seems to paint the process
around the model potentially is just
like an entropic dissipative set of
forces that the agent is trying to act
against um in some ways and that's not
necessarily a big antagonistic picture
between us and the world around us but
in some ways yeah we do have to kind of
um fight for our lives something like
that
so yeah great
points um you
could specify a generative model where

the environment was just at the exact temperature that the agent preferred or you could have one where the environment had some regions that were the preferred temperature and some that weren't and then that would lead to the question of well then how would the agent select where to move and that would take us literally exactly to expected free energy where pragmatic value

would um upweight or upregulate policy selection that led to observations aligning with preferences but again to to this top um AC action and perception certainly work

together that is a kind of way preactive Insight that that there's this um inextricable relationship between perception and action like the um uh even the opening quote of the chapter by William James is connecting thinking and and doing

[Music]

um however they're not the same we've talked about how for a given system of Interest their directions are different across a blanket um

they're they're also modeled as different in the generative model that we specify like this is looking to figure

4.3 but

perception is going to be loaded in between observations and

hidden States so perception is going to be just from a from what variables you you actually use it's it's mediating the relationship between observations and unobserved latent causes like here's the thermometer reading and then here's the estimate of what temperature it is in the room that's where perception is whereas action is seen as intervening in how hidden States change through time you're not reaching in and changing how the thermometer reading is related to temperature in the abstract you're actually doing something that modifies how temperature in the world changes through time and then the thermometer readings might be different later so it it action and perception are coming in differently to how we specify the generative

model uh Susan or or anyone else yeah so here's how I'm kind of interpreting the the planning um and this kind of brings in you the ch changing your world changing your mind versus changing your world um so changing the world if I'm if I'm trying to change my world then I'm acting on the environment and if I if it you know if there's um um and bringing in the expected ambiguity and risk that's what's

emerging and if you know if I'm not able to reach the expectations of what's changing in the world then that's going to bring me back back to changing my mind so that planning is you know is um is the the inferences that I have about

what my power in the world is and I think that you to for me that's kind of where I bring in the affordances because you the what you talked about Andrew was was I interpreted that as you someone feels

entitled um that the world's going to provide them and then they act out in the world world as if they're entitled they may well you know either have to change their mind or they're going to get reinforcement that they are entitled is that is that

cohere yeah like if there's a door and one expects to be on the other side of the door then it's like you try to open the door okay if it opens then you change the world and then you proceed on and as you described like then that was like confirmed or reinforced whereas if it doesn't open I mean what I got to get back changing my mind yeah you could keep on trying but at some point if it's not

openable there's going to have to be a belief update that that in that moment at the very

least that door can't be open otherwise it's going to be just you know you're gonna um not drink any water and you're going to die trying to open that door it's like there has to be a stopping point um and and that brings in the expected ambiguity plus

risk yes let's definitely this yes we'll look at 2.5 2.6 and see what um where these come in um

Andrew hi uh multitasking again um just

a quick
question this act action perception
conversation reminded me I have some
questions about attention and intention
do are those Concepts kind of covered or
addressed anywhere in active inference
or this
textbook attention absolutely in its
more uh technical parametric meaning of
attention as precision as well as in its
kind of broader
psychological uh meaning
intention probably more of a nebulous
concept so not to say that you couldn't
assign preferences to be intentions for
example the belief desire intention
model that's explored in live stream 46
by Ryan Smith at all where they
associate preferences just
straightforwardly um W with they talk
about the generative model in terms of
the belief desire intention folk
psychology model but in this textbook
it's um going to certainly talk about
attention and that's a major
topic cool yeah I was just noticing the
other day how um with your perceptions
there's certain modalities like your
sense Your Skin Sense and your body
sense that can be masked um in terms of
perception in a way that like noise
cannot from your conscious experience
and I thought that might say something
interesting about the active inference
model or the the structure in our
brains
yes cool yeah that this is uh
6.2 here's like different kinds of

observations so Sensor Fusion setting
and then

attention is it allocated across sensory
modalities or how is it or where is it
distributed

across um or within sensory
modalities how are those attentions
to

observations related to like Bel
unexperienced beliefs about the
world like you're not actually directly
experiencing whether there's a jumping
frog but you may experience interception
auditory visual

Jesse could you look at to Andrew's
question could you look at policy as
intention more or less because you know
it's projecting a series of actions that
you aren't you know that you'd like to
uh have happen but I was thinking that
exact same

thing so there's a quote on um page 38 I
believe

um where they're
discussing beliefs over
policies as intention and this is
related to the
expected uh free

formalism and yeah I think this is an
interesting quote to discuss if people
have any thoughts about

it nice great reference
there

I don't know if this is two two general
the statements

but it it to to go from something that
is uh Technical and within the ACT INF
onology and then kind of like zoom out

or adapt it to a more nebulous setting that's that's on you like if you choose to take that interpretation of a given aspect of active inference in a setting then it'll it'll be valid in the ways that it's valid for that setting and it won't be the whole story in the ways that it's not the whole story for that setting however that kind of a question is difficult to answer in the abstract because when we're talking about policy selection we're talking about metronomes and particles of dust and computers and groups of people so so then if but so if someone's thinking well intention I'm meaning like what I experience when I want to do something then you're already in the case of a given system of interest at which point it's no longer like a general question about the ACT INF ontology it's more about how you're adapting a term or an equation or something to a given setting at which point it'll be useful how it's useful and it'll be not the whole story in the ways that it isn't um Andrew sorry I never um unraised my hand but I I do actually have a comment on that um related to policy one thing I maybe this is discussed in later uh chapters but policy generation it seems like at least in these early equations there there a set of policies is inferred but I don't see how policies are generated is is that discussed or implied

somewhere yeah good question it connects
to the where is planning so
policies just as a first pass answer
policies are the affordances that are
available multiplied by how many uh
to to the exponent of how many time
steps the planning Horizon is so if
their affordances are up down left right
there four
affordances over one time step there's
four policies that can be selected up
down left right so for a Time Horizon of
one affordances and policies are
identical for a Time Horizon of two
you'd have 16 policies of length to up
up up down up right up left
Etc um so in this setting policies are
just
affordances over a horizon so then how
do I'm not I'm not sure that really
answers it unless affordances are
generated because it just moves the
question absolutely so how are
affordances learned or how are they
provided so like in the teamas or
something like that it's like how does
the rat know that it can move up down
left right so first we're dealing with
maps not territories so just specifying
a given Vector of affordances doesn't
mean that the rat knows that it can do
those affordances or has any kind of
like awareness uh about them it's just
us describing what we've seen the system
do so we're when we're talking about the
afford the policy prior or or the Habit
Vector the prior
on um

affordances those are something that the modeler infers not necessarily the system of Interest

itself however it still is the question of Interest well how are those affordances known and H what happens when it it up down left right and then all of a sudden up down left right and raise up you know an arm or something like that like how do new affordances come into play

we can express the question in the ACT infology

but write it out for a given system of interest and then I think you'll be well closer to the

answer but certainly you can develop nested models where like there's a um a a context shift that leads to a different set of affordances but but that's a that's um a good

question let's look at 2.5 and 2.6 and just look at their partitionings

okay so here's

2. five and they write that variational free energy so variational it just means we're using variational methods so we're doing approximate basian computation not exact Baye but approximate Baye using variational

methods it may seem an abstract concept but its nature and role it plays in active becomes apparent when decomposed into quantities that are more intuitive and familiar in cognitive

science so
three
lines are different
decompositions of variational free
energy so they differently partition the
kind of
same
piece in a few
ways the
top line is more of a statistical
physics
decomposition that is most like for
example Gibbs free energy for those with
a chemistry background
whereas the second and third
line have more to do with Statistics and
model fitting and in both cases I mean
ultimately in all cases since they're
describing the exact same quantity but
especially um in this pure statistical
second and third
decompositions it highlights that we're
fitting data but also not one to overfit
data and the family of distributions
that are being utilized are whatever are
used for Q Q is the distribution that we
control from the family that we selected
so if we say we're going to do
variational inference on temperature in
this room and we're going to use a
gaussian distribution so like a a a bell
curve
distribution that doesn't mean that the
temperature in the
room is act
changing according to a bell curve or
that the thermometer is giving us
readings according to that Q

distribution maybe it has like a long tail or maybe it has some other distribution but it means that we are going to have our hands on the knobs of Q and get to the best fitting which is to say anything wider would be underfit anything tighter would be overfit those are the knobs that we can control it might be grossly inadequate but there is a way to fit the best possible

Q to a given Y data and the way to fit the best Q to a

y is with reducing the K Divergence shown here and that is variational free minimization if you can reduce that K Divergence between the variational free energy and

surprise to zero then variational inference is identical to exact basian inference because minimizing surprise is kind of like the actual Target here if you minimize surprise you've maximized model evidence you know you're getting new thermometer readings coming in if you were minimally surprised by those readings you would have the highest model evidence you would have the best belief

so instead of trying to maximize model evidence directly with like a hill climbing approach statistical physics kind of inverts that problem and says well let's think about going downhill with a gradient

descent and here is our way to like have a tractable

incrementally improvable

approach that we know is going to be moving in the right direction to surprise minimization with variational free energy minimization which is highly optimizable again doesn't mean that it's going to get exactly here doesn't even mean it'll get close but you will be able to take incremental improving moves which is kind of shown here so that's the real time beliefs the approximate posterior q and the incoming data y That's the variational free energy and again the first decomposition is more like statistical physics free energies and the second and third decompositions are more like statistics this doesn't have anything to do with the future this is just like a real time unfolding then we get into expected free energy so expected free energy extends active inference to include a prospective form of cognition which is planning so now um rather than just the kind of ball rolling down the hill okay sometimes it jumps left sometimes it jumps right but each time its movement is basically um dictated or constrained by its local surroundings so we could talk about that ball having policy decisions or taking actions in some sort of um very like real- time unfolding way but we might not need to appeal to planning however we may be interested in cognitive models that have an explicit

planning component like deliberation or consideration of multiple decisions again this isn't to say that that system does planning just that we're going to make a cognitive model that includes the planning component um key notation is when there's a Tilda little curve curvy line over a variable that is being used to refer to like a $\tilde{S}$ a whole time series of that variable over a given time Horizon

so now we move from F of Q and Y variational free energy to G of Π so it's different in a bunch of different ways expected free energy looking to the future with these series of expected hidden States and observations that of course have not happened yet whereas variational free energy was dealing with the actual data

why we also noticed that expected free is taking in as an argument policy

priors not taking in beliefs and data directly but rather even though we see the Q and the Y but what G is doing most proximately is taking in the prior on policy so this is like a vector of all the possible

policies again if it was only one time step that's the same as the affordances if it was two time steps it would be like affordances squared taking in the prior Distribution on those policies and then sharpening or reweighting them according to

this so let's think about that up down

left right two time step settings so 16 policies so if we had no a priori habit of any of those they would all be like 116th and that Vector would look like 16 tall and each value would be 116th or if we always did policy one and two and then we very rarely did the other ones it would be like 0.5 point5 and all zeros or you might put that in a computer as like 04949 and then very small values for things that are rarely done G takes in that policy prior and it updates it according to this calculation and so for people who want to look at the code that's very helpful to see where exactly this comes into play takes in that vector and then it sharpens or updates it to give more weight make the policy posterior bigger where where the policy contributes Information Gain or price Matic value pragmatic value Rewards or updates or increases the waiting on policies that bring expected future observations into alignment with preferences that is the definition of pragmatic value the definition of Information Gain or epistemic value is about the difference that you're expected to um get depending on the observations that you expect to see so it's definitely a

lot however that is what this
top and most common decomposition of
expected free energy
does and then as Susan brought up
another decomposition of
G is rather than thinking about what G
does as upweighting based upon epistemic
impracticable value you could think about
G as changing the The Waiting of
policies based upon their expected
ambiguities and
risk um David and then
Andrew hi I guess a
major a major obstacle for me to
understand all of this is literally the
mathematic construction of this sentence
so like I I for example here I just
don't know where the what the t_i means
in X and Y or what where the C comes
from just ask me because I don't know if
somebody has this sense of
discomfort not being able to decode what
what's this trying to tell me and I
don't know if somebody have some good
resource on how to really understand
this this
construction yeah thanks for asking
totally agreed
um hopefully you're in a good resource
now and thank you again for
asking depending on someone's
familiarity with math or computer
science and programming there's
definitely a lot of approaches but the
big approach is many coats of paint like
coarse grain it just say Okay 2.6 is
expected free energy and you can go down
fractal rabbit holes for the rest of

your life or at some point you can just say Okay expected for free energy is about updating beliefs about action another really helpful resource that again I encourage people to contribute to is the natural language descriptions of the questions it's a lot easier to read there's still a lot of terms in here but the blue terms are ones where we have Mouse over definitions um and then also maybe someone's interested in playing around more with the AI explanations that we used these may be helpful um certainly on your own notebook or however like the notation uh and understanding like you know Q approximate posterior x hidden States Through Time y observations Through Time π policy it is like learning a vocabulary or a symbol matching exercise um and math is a language and acin is a dialect so it just is a lot of exposure and hearing about it Kristoff they pull the equation out of the Hat yeah in in in some ways yes also the textbook is uh is distilling and summarizing across research and there are more lengthy derivations elsewhere the the math learning group again to the extent that people want to to do these activities um there's always been like a

more fundamentals of math learning group
side like just like what does it mean
when we have two matrices and then we
just like put them next to each other
like what what does that mean to
multiply them and then also some people
like Jonathan and Ali and others have
worked on the derivations of the
equations and sanj N josi's textbook and
other resources help so it's just like
but being in the game is the key
enabling step um

Andrew yeah um actually kind of related
to this uh just one toide for some of
the Greek characters as soon as I start
like memorizing what the pronunciation
of the characters are you find that
there is a mapping between like me and
mind there's some alliteration going on
um so that might be a little pneumatic
to help um the other thing is I was
wondering with the assignment of like um
epistemic value to part of these
equations is that a generally agreed
upon format Within in the sort of
scientific body or was that sort of
arbitrarily mapped by the researchers um
putting these equations
together so so like those statistical
yeah um beian sort of uh equations is
that like agreed upon as um denoting
that qualitative uh term
in the case of variational free energy
yes I'd say it's absolutely non-c
contentious variational basian methods
or energy based methods with these
decompositions have been in play for
decades expected free energy extends

vfe

into the discussion of observations that
have not happened

yet also these are non-c contentious
decompositions I believe I don't know I
don't have any survey data on on who
knows or believes what but but I don't
think that these are
contentious about the latch there is a
column

here um if you want the raw
latch that is a a good
technique you can paste this into yeah
gp4 can use images as
input um this is why the stigar G is so
important the notes that people write in
the questions like the more that people
add the more material there are for
human and non-human

Learners um and we make do and we stay
in the game with what we have like three
or six months ago when we were
converting the m to latch this was the
best that we could

do and it works really well with Cod AI
Cod AI does not have the image
recognition yet but if slash when they
do we'll be able to add
another column and just say Okay given
what's in the

figure dot dot dot dot
dot so yeah lot of different interesting
emerging ways

to work with the
material cool so that is
um chapter

two on the low
road for the next two

weeks we'll head into chapter 3 on the h
so thank you everyone see you next
time thank you thank you bye thank you
this is great